CHAPTER

21

Filter Comparison

Decisions, decisions, decisions! With all these filters to choose from, how do you know which to use? This chapter is a head-to-head competition between filters; we'll select champions from each side and let them fight it out. In the first match, *digital* filters are pitted against *analog* filters to see which technology is best. In the second round, the windowed-sinc is matched against the Chebyshev to find the king of the *frequency domain* filters. In the final battle, the moving average fights the single pole filter for the *time domain* championship. Enough talk; let the competition begin!

Match #1: Analog vs. Digital Filters

Most digital signals originate in analog electronics. If the signal needs to be filtered, is it better to use an analog filter before digitization, or a digital filter after? We will answer this question by letting two of the best contenders deliver their blows.

The goal will be to provide a low-pass filter at 1 kHz. Fighting for the analog side is a six pole Chebyshev filter with 0.5 dB (6%) ripple. As described in Chapter 3, this can be constructed with 3 op amps, 12 resistors, and 6 capacitors. In the digital corner, the windowed-sinc is warming up and ready to fight. The analog signal is digitized at a 10 kHz sampling rate, making the cutoff frequency 0.1 on the digital frequency scale. The length of the windowed-sinc will be chosen to be 129 points, providing the same 90% to 10% roll-off as the analog filter. Fair is fair. Figure 21-1 shows the frequency and step responses for these two filters.

Let's compare the two filters blow-by-blow. As shown in (a) and (b), the analog filter has a 6% ripple in the passband, while the digital filter is perfectly flat (within 0.02%). The analog designer might argue that the ripple can be *selected* in the design; however, this misses the point. The flatness achievable with analog filters is limited by the accuracy of their resistors and